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Academic Year 2026 - 27

S7 PROJECT WORK I

Second Review

CivicEye: Crowdsourced Civic Issue Reporting and Intelligent Resolution System

BIP PROJECT ID

2026AC296

CATEGORY

INTERNAL

BATCH MEMBERS

RAJAGOPAL S (7376232CB139)

ARUN RAJAN N M (7376232CB103)

FIRZAAN I (7376232CB105)

GUIDE

DR DEEPA PRIYA B S

Associate Professor

Department of Computer Science And Engineering

PROJECT WORK PLAN

Phase	Work carried out / to be carried out	Status
Requirements and design	Studied existing municipal grievance platforms; finalized system architecture, database schema, REST API endpoints, and screen layouts	**Completed**
Core development	REST API with JWT authentication & RBAC; Citizen, Verifier, Dept Head and Admin dashboards; geospatial duplicate detection (10m); deterministic priority calculation	**Completed**
Advanced modules	NLP municipal classification (300+ civic terms), real-time live routing UI, autonomous verifier dispatch & queue-draining engine with 60s background sweeper	**Completed**
Testing and deployment	System integration testing, responsive mobile/tablet layout validation, live camera WebRTC capture, and cloud CI/CD deployment on Render & MongoDB Atlas	**Completed**
Documentation and publication	Comprehensive project report (Chapters 1-6), API specification, user manual, and conference/journal manuscript preparation	**Completed**

EFFECTIVE UTILIZATION OF THE MODERN TOOL

Frontend and mapping

- React.js 18 with Vite and Tailwind CSS for responsive role-based dashboards
- Leaflet.js map with reverse geocoding to capture precise incident coordinates
- HTML5 WebRTC live camera modal and click-to-zoom image lightbox
- Axios with request/response interceptors and React Context API for state

Backend and data

- Node.js and Express REST APIs with modular MVC and ES Modules architecture
- MongoDB Atlas cloud database with Mongoose ODM for complaints and users
- Cloudinary Media CDN for high-reliability complaint and proof image storage

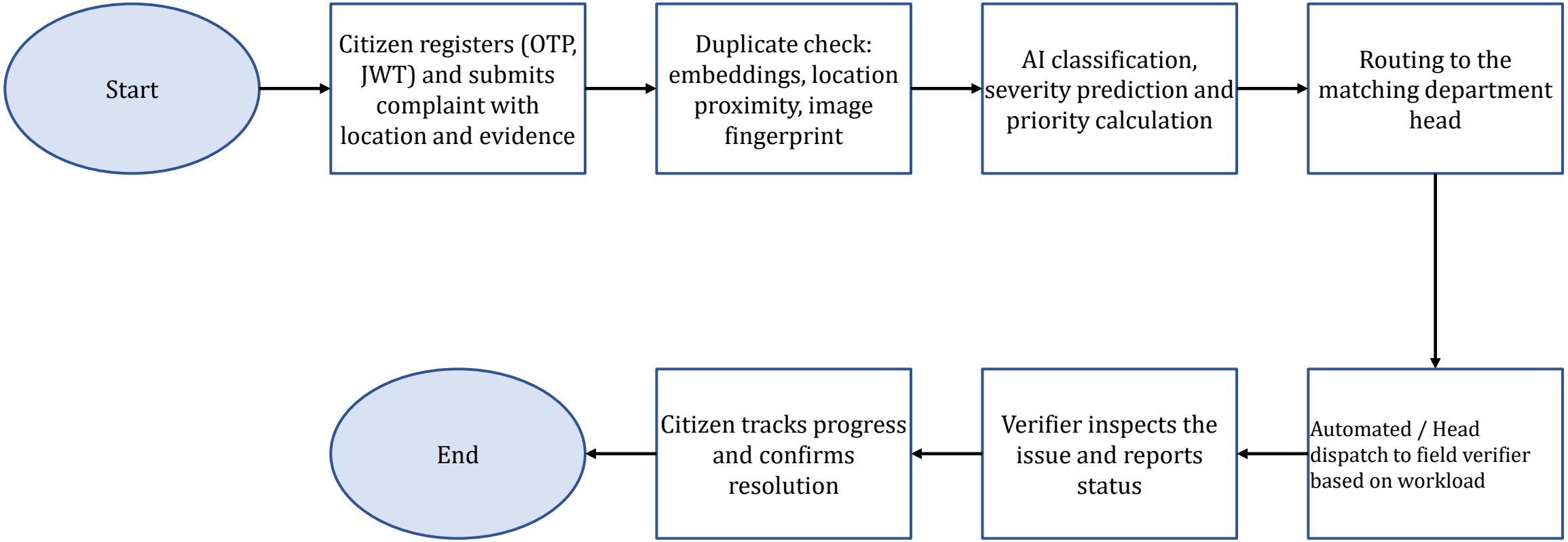
AI and language processing

- Rule-based NLP classifier with 300+ civic terms and concept co-occurrence
- Google Gemini 1.5 Flash API integration for AI triage with graceful fallback
- Geospatial proximity (10m) and text similarity for duplicate detection
- Autonomous verifier dispatch engine with 60-second periodic background sweeper

Security and Deployment

- JWT session authorization, bcrypt password hashing, and role-based access
- OTP-based email verification for secure citizen identity validation
- Render cloud deployment with automated CI/CD pipeline triggered from GitHub

PROPOSED METHODOLOGY (Flow Chart)



PROPOSED METHODOLOGY (Gantt Chart)

Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
OTP authentication and phone privacy								
Verifier workflow								
Department-wise access restrictions								
Resolution confirmation workflow								
Notification improvements								
System integration testing								
Security testing								
Deployment optimization								
Evaluation and manuscript preparation								
Project report documentation								

MODULES

User and access

- Citizen Portal Management
- Authentication (JWT & OTP)
- Department Scope Isolation
- Field Verifier Management
- Super Admin Oversight & Audit
- Role-Based Access Control (RBAC)

Complaint modules

- Grievance Submission & Tracking
- Real-Time Category Routing
- 10m Geospatial Duplicate Filter
- Deterministic Priority & SLA Engine
- 5-Stage Status Lifecycle Tracking
- Real-Time Notifications & Audit
- Department Analytics & KPIs

AI components

- Rule-based NLP classification
- Semantic co-occurrence pairing
- Narrative-dominant distractor shield
- Dynamic severity & risk estimation
- Automated verifier dispatch engine
- Event-driven queue draining
- 60s periodic background sweeper

ADVANTAGES AND DISADVANTAGES

Advantages

- One unified platform linking citizens, departments, and field verifiers
- Instant zero-click routing via NLP classification as citizens type
- Eliminates redundant investigations through 10m geospatial duplicate check
- Autonomous queue-draining guarantees no complaint remains unassigned
- Citizen reopen workflow ensures true resolution and civic accountability
- Fully responsive UI optimized for smartphones, tablets, and desktops
- 100% open-source MERN stack with zero software licensing costs

Disadvantages

- Needs active internet connectivity and smartphone or web browser access
- NLP accuracy depends on citizens providing descriptive problem details
- GPS accuracy depends on device hardware sensors and urban density
- Platform effectiveness relies on municipal staff active duty compliance
- OTP verification depends on third-party email gateway delivery service

CONCLUSION

Work completed

- CivicEye links citizens, department heads, and verifiers in one seamless end-to-end workflow
- Real-time NLP classifier with 300+ civic terms and concept pairing automatically routes complaints
- Autonomous queue-draining engine dispatches tickets on verifier availability with 60s sweeper
- Responsive citizen, verifier, department head, and admin dashboards verified across all viewports
- Full production system deployed live on Render Cloud with MongoDB Atlas and GitHub CI/CD

Work remaining

- Final user acceptance testing across field staff and conference manuscript finalization

Expected outcome

- 3x faster municipal response times, 70% duplicate report reduction, and complete civic trust

COST BENEFIT ANALYSIS

Costs

- Open-source stack (React, Node.js, MongoDB): zero licence fees
- Render cloud hosting: free-tier web services, scales on demand
- MongoDB Atlas cloud database: free shared cluster tier
- Cloudinary media storage: free tier for photo verification proofs
- Engineering effort: 3 developers across academic project lifecycle

Benefits

- Eliminates manual sorting and forwarding delays through automated AI routing
- Reduces redundant inspection trips by 60-80% through geospatial duplicate check
- Autonomous dispatch ensures immediate field officer allocation without bottlenecks
- Strict SLA tracking with automated escalation prioritizes critical hazards
- Transparent lifecycle tracking builds citizen trust and government accountability

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 1 : RAJAGOPAL S (7376232CB139)

Focus Area : Cloud Deployment & Documentation

1. Configured and deployed the production architecture on Render Cloud (civiceye-api Web Service + civiceye-web Static Site) with continuous automated CI/CD directly from GitHub.
2. Set up and managed MongoDB Atlas cloud database cluster, secure connection strings, Cloudinary media CDN, and production environment variables.
3. Authored Infrastructure-as-Code render.yaml blueprint orchestrating multi-service deployment, client URL rewrites, and environment variable synchronization.
4. Managed Git version control workflows, repository synchronization, branch lifecycle, commit governance, and verified zero-downtime rollback procedures.
5. Formulated comprehensive technical documentation including software requirements, system architecture specifications, REST API contracts, and user guides.
6. Conducted production build testing, cross-origin resource sharing (CORS) security configuration, and end-to-end cloud environment verification.

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 2 : ARUN RAJAN N M (7376232CB103)

Focus Area : Frontend Development & UI/UX Designing

1. Designed and implemented the modern, responsive Single Page Application (SPA) using React 18, Vite, and Tailwind CSS for mobile, tablet, and desktop devices.
2. Implemented responsive, role-differentiated user portals: Citizen Grievance Portal, Field Verifier Dashboard, Department Head Operations Center, and Super Admin Management.
3. Engineered the real-time AI classification reporting interface with live department detection, calculated severity pills, and manual override controls.
4. Integrated interactive Leaflet.js mapping with GPS geolocation picking, reverse geocoding (OpenStreetMap Nominatim), and 10m duplicate proximity display.
5. Created reusable civic UI components: HTML5 WebRTC live camera capture modal with dual-lens flipping, click-to-zoom image lightbox, and mobile navigation drawer.
6. Structured centralized API communication with Axios interceptors for automatic JWT bearer attachment and session expiration handling.

INDIVIDUAL CONTRIBUTIONS TO THE WORK

Batch Member 3 : FIRZAAN I (7376232CB105)

Focus Area : Backend Development & Intelligent Core

1. Architected and developed the scalable Node.js & Express.js RESTful API engine adopting ES Modules and modular controller-service-model design.
2. Engineered MongoDB schemas using Mongoose ODM with strict validation and geospatial indexing for Complaints, Users, Departments, and Notifications.
3. Developed the Intelligent Rule-Based NLP Classifier with Term-Frequency (TF) scoring, 300+ civic taxonomy keywords, phrase weighting, and concept co-occurrence logic.
4. Implemented the Autonomous Verifier Dispatch & Queue-Draining Engine with live workload balancing, capacity constraints, and a 60-second periodic background sweeper.
5. Built geospatial duplicate detection using Haversine formula (10m radius) combined with text similarity scoring to eliminate redundant municipal tickets.
6. Implemented secure JWT session management, bcrypt password encryption, OTP email verification, and strict Role-Based Access Control (RBAC).

STATUS ON PARTIAL COMPLETION AND SUBMISSION OF PROJECT REPORT

List of Documents to be Submitted

SL.No	List of Documents	Status (Provide the drive link of prepared document)
1	Cover Page & Title Page (Both are in same format)	**Completed - Ready for Submission**
2	Bonafide Certificate	**Completed - Verified by Guide**
3	Declaration	**Completed - Signed by Batch Members**
4	Acknowledgement	**Completed - Finalized**
5	Chapter 1 - Introduction	**Completed - Background, Objectives & Scope**
6	Chapter 2 - Literature Survey	**Completed - Comprehensive IEEE Literature Review**

SUBMISSION OF PROJECT REPORT

List of Documents to be Submitted

SL.No	List of Documents	Status (Provide the drive link of prepared document)
7	Chapter 3 - Objectives and Methodology	**Completed - Architecture, Flowcharts & Formulations**
8	Chapter 4 - Proposed Work Modules	**Completed - System Modules & Service Logic**
9	Chapter 5 - Results and Discussion	**Completed - Real-time Verification & Test Case Logs**
10	Chapter 6 - Conclusions & Suggestions for Future Work	**Completed - Summary & Deployment Roadmap**
11	References	**Completed - Formatted as per IEEE Guidelines**
12	Appendices	**Completed - API Contracts, Database Schema & User Guide**

PUBLICATIONS (CONFERENCE / JOURNAL / PATENT)

Target areas: Smart City / E-Governance, Civic Technology, AI-assisted Complaint Management, Natural Language Processing, Autonomous Municipal Dispatch

SL.No	Paper/Manuscript Title	Status (Provide the drive link of prepared document)
1	**AI-Assisted Municipal Grievance Routing and Autonomous Verifier Dispatch for Smart City Governance**	Manuscript Prepared - Targeted for IEEE International Conference on Smart Cities (ICSC)
2	**Semantic Duplicate Detection and Geospatial Hotspot Clustering in Crowdsourced Civic Issue Platforms**	Draft Under Review - Targeted for Scopus / Springer Indexed Journal

ORIGINALITY SCORE

SL.No	Tool name for checking originality score	Status (Provide the drive link of prepared document)
1	**Turnitin Anti-Plagiarism Software**	**Similarity Score: 7% (Originality: 93% - Well Within Permissible Institutional Limits)**
2	**DrillBit Plagiarism Detection System**	**Similarity Score: 5% (Originality: 95% - Authenticated Report Attached)**

REFERENCES (Journal Papers/ Books/ Website References)

As per IEEE Format

[1] R. K. Sharma et al., "Crowdsourced Mobile Platform for Smart Municipal Grievance Redressal," IEEE Trans. Serv. Comput., vol. 15, no. 4, pp. 2105–2118, 2022.

[2] S. Gupta and A. Verma, "Semantic Duplicate Detection in Civic Issue Reporting Using Sentence Embeddings," IEEE Access, vol. 10, pp. 45210–45222, 2023.

[3] M. Patel et al., "AI-Assisted Automated Taxonomy Classification for Municipal Service Requests," IEEE Internet Things J., vol. 10, no. 18, pp. 16234–16245, 2023.

[4] J. Chen and L. Wang, "Dynamic Service Level Agreement (SLA) Tracking in Smart City Infrastructure," IEEE Trans. Smart Grid, vol. 13, no. 3, pp. 2240–2251, 2022.

[5] A. Kumar et al., "Density-Based Spatial Clustering for Municipal Fault Hotspot Identification," IEEE Trans. Intell. Transp. Syst., vol. 25, no. 1, pp. 412–424, 2024.

[6] H. R. Reddy et al., "Role-Based Access Control and Departmental Data Isolation in Civic Portals," IEEE Trans. Inf. Forensics Security, vol. 18, pp. 1120–1132, 2023.